

Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Worked Solutions

Coordinate Geometry Basics

Q001 **Written****A (−4), B (8): distance AB****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Q002 **Written****A (−5), B (7): distance AB****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Worked Solutions

Coordinate Geometry Basics

Q003 Written

For two points A (-2) and B (6) , find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Q004 Written

For two points A (4) and B (10) , find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 6.
3. Final answer: 6.

Final answer

6

Worked Solutions

Coordinate Geometry Basics

Q005 MCQ

Find a (-4) , B (8) : distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C**Final answer**

12

Q006 MCQ

Find a (-5) , B (7) : distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C**Final answer**

12

Worked Solutions

Coordinate Geometry Basics

Q007 MCQ

For two points A (-2) and B (6) , find the distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: C**Final answer**

8

Q008 MCQ

For two points A (4) and B (10) , find the distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 6.

Correct option: C**Final answer**

6

Worked Solutions

Coordinate Geometry Basics

Q009 Written

A (−4), B (8): internal point C in ratio 2:1**Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q010 Written

A (−5), B (7): internal point C in ratio 2:1**Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 3.
3. Final answer: 3.

Final answer

3

Worked Solutions

Coordinate Geometry Basics

Q011 Written

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q012 Written

For two points A (4) and B (10) , find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Worked Solutions

Coordinate Geometry Basics

Q013 MCQ

Find a (-4) , B (8) : internal point C in ratio 2:1:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B**Final answer**

4

Q014 MCQ

Find a (-5) , B (7) : internal point C in ratio 2:1:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 3.

Correct option: C**Final answer**

3

Worked Solutions

Coordinate Geometry Basics

Q015 MCQ

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B**Final answer**

4

Q016 MCQ

For two points A (4) and B (10) , find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: B**Final answer**

8

Worked Solutions

Coordinate Geometry Basics

Q017 **Written****A (−4), B (8): external point D in ratio 2:3****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -28 .
3. Final answer: -28 .

Final answer -28 Q018 **Written****A (−5), B (7): external point D in ratio 2:5****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -13 .
3. Final answer: -13 .

Final answer -13

Worked Solutions

Coordinate Geometry Basics

Q019 Written

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 18.
3. Final answer: 18.

Final answer

18

Q020 Written

For two points A (4) and B (10) , find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions

Coordinate Geometry Basics

Q021 MCQ

Find a (-4) , B (8) : external point D in ratio 2:3:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -28 .

Correct option: A**Final answer** -28

Q022 MCQ

Find a (-5) , B (7) : external point D in ratio 2:5:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -13 .

Correct option: B**Final answer** -13

Worked Solutions

Coordinate Geometry Basics

Q023 MCQ

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 18.

Correct option: A**Final answer**

18

Q024 MCQ

For two points A (4) and B (10) , find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: B**Final answer**

1

Worked Solutions

Coordinate Geometry Basics

Q025 **Written****A (−4), B (8): midpoint M****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q026 **Written****A (−5), B (7): midpoint M****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions

Coordinate Geometry Basics

Q027 **Written****For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB.****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q028 **Written****For two points A (4) and B (10) , find the coordinate of midpoint M of segment AB.****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 7.
3. Final answer: 7.

Final answer

7

Worked Solutions

Coordinate Geometry Basics

Q029 MCQ

Find a (-4) , B (8) : midpoint M:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: C**Final answer**

2

Q030 MCQ

Find a (-5) , B (7) : midpoint M:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: A**Final answer**

1

Worked Solutions

Coordinate Geometry Basics

Q031 MCQ

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: D**Final answer**

2

Q032 MCQ

For two points A (4) and B (10) , find the coordinate of midpoint M of segment AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 7.

Correct option: A**Final answer**

7

Worked Solutions

Coordinate Geometry Basics

Q033 Challenge**Pods A (-8) and B (-3) : fibre-optic tether length****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is 5.

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Q034 Challenge**Pods A (-8) , B (-3) : internal sensor point in ratio 1:4****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -7 .

Final answer -7

Worked Solutions

Coordinate Geometry Basics

Q035 Challenge**Pods A (-8) , B (-3) : external outpost in ratio 3:4****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -23 .

Final answer -23

Worked Solutions**Coordinate Geometry Basics****Q036 Challenge****Pods A (−8), B (−3): midpoint supply hub****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $-11/2$.

Final answer $-11/2$

Worked Solutions

Coordinate Geometry Basics

Q037 Challenge

Find pods A (-8) and B (-3) : fibre-optic tether length:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: A

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Q038 Challenge

Find pods A (-8) , B (-3) : internal sensor point in ratio 1:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -7 .

Correct option: C

Final answer

-7

Worked Solutions

Coordinate Geometry Basics

Q039 Challenge

Find pods A (-8) , B (-3) : external outpost in ratio 3:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -23 .

Correct option: B

Final answer

-23

Worked Solutions

Coordinate Geometry Basics

Q040 Challenge**Find pods A (-8) , B (-3) : midpoint supply hub:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $-11/2$.

Correct option: C**Final answer** $-11/2$

Worked Solutions

Coordinate Geometry Basics

Q041 **Written****Point a has coordinates (2,3): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*Q042 **Written****Point b has coordinates (2,-3): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q043 **Written****Point c has coordinates $(-2,3)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q044 **Written****Point d has coordinates $(-2,-3)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q045 **Written****Point a has coordinates $(-4,2)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q046 **Written****Point b has coordinates $(1,2)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*

Worked Solutions

Coordinate Geometry Basics

Q047 **Written****Point c has coordinates (3,-2): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*Q048 **Written****Point d has coordinates (-3,-1): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q049 **Written****Point a has coordinates $(-3,1)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q050 **Written****Point b has coordinates $(2,-8)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q051 **Written****Point c has coordinates (4,4): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*Q052 **Written****Point d has coordinates (-6,-4): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q053 Written

Point e has coordinates $(-2,6)$: determine its quadrant**Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*

Q054 Written

Point f has coordinates $(3,-3)$: determine its quadrant**Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q055 **Written****Point g has coordinates $(-5, -7)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*Q056 **Written****Point h has coordinates $(1, 4)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*

Worked Solutions

Coordinate Geometry Basics

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: C**Final answer***I*

Q058 MCQ

Point b has coordinates (2,-3): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: B**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q059 MCQ

Point c has coordinates $(-2,3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A**Final answer***II*

Q060 MCQ

Point d has coordinates $(-2,-3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q061 MCQ

Point a has coordinates $(-4,2)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A**Final answer***II*

Q062 MCQ

Point b has coordinates $(1,2)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: C**Final answer***I*

Worked Solutions

Coordinate Geometry Basics

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: B**Final answer***IV*

Q064 MCQ

Point d has coordinates (-3,-1): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: A**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q065 MCQ

Point a has coordinates $(-3,1)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: B**Final answer***II*

Q066 MCQ

Point b has coordinates $(2,-8)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: A**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: A**Final answer***I*

Q068 MCQ

Point d has coordinates (-6,-4): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q069 MCQ

Point e has coordinates $(-2,6)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: B**Final answer***II*

Q070 MCQ

Point f has coordinates $(3,-3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: C**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q071 MCQ

Point g has coordinates $(-5,-7)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Q072 MCQ

Point h has coordinates $(1,4)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: D**Final answer***I*

Worked Solutions

Coordinate Geometry Basics

Q073 Written

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3, -2)$.

Final answer

$(-3, -2)$

Q074 Written

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, -2)$.

Final answer

$(-1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q075 Written

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-4, 2)$.

Final answer

$(-4, 2)$

Q076 Written

For point A $(1, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, 2)$.

Final answer

$(1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q077 Written

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3, -4)$.

Final answer

$(-3, -4)$

Q078 MCQ

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3, -2)$.

Correct option: C

Final answer

$(-3, -2)$

Worked Solutions

Coordinate Geometry Basics

Q079 MCQ

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, -2)$.

Correct option: D

Final answer

$(-1, -2)$

Q080 MCQ

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-4, 2)$.

Correct option: C

Final answer

$(-4, 2)$

Worked Solutions

Coordinate Geometry Basics

Q081 MCQ

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (1, 2).

Correct option: D

Final answer

(1, 2)

Q082 MCQ

For point A (-3, 4), find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-3, -4).

Correct option: C

Final answer

(-3, -4)

Worked Solutions

Coordinate Geometry Basics

Q083 Written

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, 2)$.

Final answer

$(3, 2)$

Q084 Written

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, 2)$.

Final answer

$(1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q085 Written

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4, -2)$.

Final answer

$(4, -2)$

Q086 Written

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, -2)$.

Final answer

$(-1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q087 Written

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, 4)$.

Final answer

$(3, 4)$

Q088 MCQ

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 2)$.

Correct option: C

Final answer

$(3, 2)$

Worked Solutions

Coordinate Geometry Basics

Q089 MCQ

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1, 2)$.

Correct option: C**Final answer** $(1, 2)$

Q090 MCQ

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4, -2)$.

Correct option: A**Final answer** $(4, -2)$

Worked Solutions

Coordinate Geometry Basics

Q091 MCQ

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, -2)$.

Correct option: A**Final answer** $(-1, -2)$

Q092 MCQ

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 4)$.

Correct option: A**Final answer** $(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q093 **Written****For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, -2)$.

Final answer $(3, -2)$ Q094 **Written****For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, -2)$.

Final answer $(1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q095 Written

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4, 2)$.

Final answer

$(4, 2)$

Q096 Written

For point A $(1, -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, 2)$.

Final answer

$(-1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q097 **Written****For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, -4)$.

Final answer $(3, -4)$ Q098 **MCQ****For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, -2)$.

Correct option: B**Final answer** $(3, -2)$

Worked Solutions

Coordinate Geometry Basics

Q099 MCQ

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1, -2)$.

Correct option: B

Final answer

$(1, -2)$

Q100 MCQ

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4, 2)$.

Correct option: B

Final answer

$(4, 2)$

Worked Solutions

Coordinate Geometry Basics

Q101 MCQ

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-1, 2).

Correct option: A**Final answer**

(-1, 2)

Q102 MCQ

For point A (-3, 4), find the coordinates of point R symmetric with respect to the origin:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (3, -4).

Correct option: D**Final answer**

(3, -4)

Worked Solutions

Coordinate Geometry Basics

Q103 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(-3, -4)$.

Final answer

$(-3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q104 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3, 4)$.

Final answer

$(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q105 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3, -4)$.

Final answer

$(3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q106 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3, -4)$.

Correct option: A

Final answer

$(-3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q107 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 4)$.

Correct option: B

Final answer

$(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q108 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, -4)$.

Correct option: C

Final answer

$(3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q109 Written

Distance between A (2, 6) and B (5, 10)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

5

Q110 Written

Distance between A (4, 3) and B (1, 2)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{10}$.

Final answer $\sqrt{10}$

Worked Solutions

Coordinate Geometry Basics

Q111 Written

Distance between A($\sqrt{3}$, -1) and B (0, 1)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{7}$.

Final answer

$$\sqrt{7}$$

Q112 Written

Distance between A (1, 2) and B (11, 7)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $5\sqrt{5}$.

Final answer

$$5\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q113 Written

Distance between A (3, -1) and B (-1, 3)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $4\sqrt{2}$.

Final answer

$$4\sqrt{2}$$

Q114 MCQ

Find distance between A (2, 6) and B (5, 10):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C**Final answer**

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q115 MCQ

Find distance between A (4, 3) and B (1, 2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A**Final answer**

$$\sqrt{10}$$

Q116 MCQ

Find distance between A($\sqrt{3}$, -1) and B (0, 1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{7}$.

Correct option: C**Final answer**

$$\sqrt{7}$$

Worked Solutions

Coordinate Geometry Basics

Q117 MCQ

Find distance between A (1, 2) and B (11, 7):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $5\sqrt{5}$.

Correct option: B**Final answer**

$$5\sqrt{5}$$

Q118 MCQ

Find distance between A (3, -1) and B (-1, 3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $4\sqrt{2}$.

Correct option: A**Final answer**

$$4\sqrt{2}$$

Worked Solutions

Coordinate Geometry Basics

Q119 Written

A (3, -2), **B** (x, 1), **distance 5: find x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Q120 Written

A (-1, 3), **B** (x, 1), **distance $\sqrt{5}$: find x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -2, 0$.

Final answer

$$x = -2, 0$$

Worked Solutions

Coordinate Geometry Basics

Q121 Written

A (3, -1), B (x, 2), distance 5: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Q122 Written

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $y = -5, -1$.

Final answer

$$y = -5, -1$$

Worked Solutions

Coordinate Geometry Basics

Q123 Written

A $(-4, -2)$, B $(x, 3)$, distance $5\sqrt{5}$: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -14, 6$.

Final answer

$$x = -14, 6$$

Q124 MCQ

A $(3, -2)$, B $(x, 1)$, distance 5: find x :

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1, 7$.

Correct option: B

Final answer

$$x = -1, 7$$

Worked Solutions

Coordinate Geometry Basics

Q125 MCQ

A $(-1, 3)$, **B** $(x, 1)$, **distance** $\sqrt{5}$: **find x:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2, 0$.

Correct option: C**Final answer**

$$x = -2, 0$$

Q126 MCQ

A $(3, -1)$, **B** $(x, 2)$, **distance** 5: **find x:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1, 7$.

Correct option: B**Final answer**

$$x = -1, 7$$

Worked Solutions

Coordinate Geometry Basics

Q127 MCQ

A (1, -3), **B** (-2, y), **distance** $\sqrt{13}$: **find** y :**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $y = -5, -1$.

Correct option: A**Final answer**

$$y = -5, -1$$

Q128 MCQ

A (-4, -2), **B** (x , 3), **distance** $5\sqrt{5}$: **find** x :**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -14, 6$.

Correct option: D**Final answer**

$$x = -14, 6$$

Worked Solutions

Coordinate Geometry Basics

Q129 Written

Distance from origin to A (2, -4)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $2\sqrt{5}$.

Final answer

$$2\sqrt{5}$$

Q130 Written

Distance from origin to A (-3, 4)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q131 MCQ

Find distance from origin to A (2, -4):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: C**Final answer**

$$2\sqrt{5}$$

Q132 MCQ

Find distance from origin to A (-3, 4):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C**Final answer**

$$5$$

Worked Solutions**Coordinate Geometry Basics****Q133 Challenge**

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $x = -2, 3$.

Final answer

$$x = -2, 3$$

Worked Solutions

Coordinate Geometry Basics

Q134 Challenge

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x :

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2, 3$.

Correct option: C

Final answer

$$x = -2, 3$$

Worked Solutions

Coordinate Geometry Basics

Q135 **Written****Point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, 0)$.

Final answer $(2, 0)$ Q136 **Written****Point on the y-axis equidistant from A $(4, 2)$ and B $(1, -1)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, 3)$.

Final answer $(0, 3)$

Worked Solutions

Coordinate Geometry Basics

Q137 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, 0)$.

Final answer $(2, 0)$ Q138 **Written****Point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, 5/6)$.

Final answer $(0, 5/6)$

Worked Solutions

Coordinate Geometry Basics

Q139 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(17/2, 0)$.

Final answer $(17/2, 0)$ Q140 **MCQ****Find point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 0)$.

Correct option: C**Final answer** $(2, 0)$

Worked Solutions

Coordinate Geometry Basics

Q141 MCQ

Find point on the y-axis equidistant from A (4, 2) and B (1, -1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (0, 3).

Correct option: A**Final answer**

(0, 3)

Q142 MCQ

Find point on the x-axis equidistant from A (-1, 2) and B (4, 3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, 0).

Correct option: C**Final answer**

(2, 0)

Worked Solutions

Coordinate Geometry Basics

Q143 MCQ

Find point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 5/6)$.

Correct option: C**Final answer** $(0, 5/6)$

Q144 MCQ

Find point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(17/2, 0)$.

Correct option: B**Final answer** $(17/2, 0)$

Worked Solutions

Coordinate Geometry Basics

Q145 **Written****Point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(13/8, 9/4)$.

Final answer $(13/8, 9/4)$ Q146 **Written****Point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(-4, -11)$.

Final answer $(-4, -11)$

Worked Solutions

Coordinate Geometry Basics

Q147 **Written****Point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (2, 1).

Final answer

(2, 1)

Q148 **Written****Point on $y = x - 1$ equidistant from A (-1, 5) and B (7, -1)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (3, 2).

Final answer

(3, 2)

Worked Solutions

Coordinate Geometry Basics

Q149 **Written****Point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (−2, 3).

Final answer

(−2, 3)

Q150 **MCQ****Find point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1):****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (13/8, 9/4).

Correct option: B**Final answer**

(13/8, 9/4)

Worked Solutions

Coordinate Geometry Basics

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-4, -11).

Correct option: C**Final answer**

(-4, -11)

Q152 MCQ

Find point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, 1).

Correct option: A**Final answer**

(2, 1)

Worked Solutions

Coordinate Geometry Basics

Q153 MCQ

Find point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 2)$.

Correct option: C**Final answer** $(3, 2)$

Q154 MCQ

Find point on $y = -x + 1$ equidistant from A $(1, 2)$ and B $(-3, 6)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2, 3)$.

Correct option: B**Final answer** $(-2, 3)$

Worked Solutions

Coordinate Geometry Basics

Q155 Challenge

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $3\sqrt{5}$.

Final answer

$$3\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q156 Challenge

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $3\sqrt{5}$.

Correct option: A

Final answer

$$3\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q157 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(4/3, 11/3)$.

Final answer $(4/3, 11/3)$

Worked Solutions

Coordinate Geometry Basics

Q158 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-15/4, 13/4)$.

Final answer $(-15/4, 13/4)$

Worked Solutions

Coordinate Geometry Basics

Q159 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (2, -9/5).

Final answer

(2, -9/5)

Worked Solutions

Coordinate Geometry Basics

Q160 Written

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3, 3)$.

Final answer

$(3, 3)$

Worked Solutions

Coordinate Geometry Basics

Q161 Written

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2/3, -5/3)$.

Final answer

$(-2/3, -5/3)$

Worked Solutions

Coordinate Geometry Basics

Q162 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3, 11/3)$.

Correct option: C**Final answer** $(4/3, 11/3)$

Worked Solutions

Coordinate Geometry Basics

Q163 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-15/4, 13/4)$.

Correct option: C

Final answer

$(-15/4, 13/4)$

Worked Solutions

Coordinate Geometry Basics

Q164 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, -9/5).

Correct option: C

Final answer

(2, -9/5)

Worked Solutions

Coordinate Geometry Basics

Q165 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 3)$.

Correct option: C

Final answer

$(3, 3)$

Worked Solutions

Coordinate Geometry Basics

Q166 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2/3, -5/3)$.

Correct option: A

Final answer

$(-2/3, -5/3)$

Worked Solutions

Coordinate Geometry Basics

Q167 Written

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, 19/2)$.

Final answer

$(2, 19/2)$

Worked Solutions

Coordinate Geometry Basics

Q168 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(11, -18)$.

Final answer $(11, -18)$

Worked Solutions

Coordinate Geometry Basics

Q169 Written

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (10, -15).

Final answer

(10, -15)

Worked Solutions

Coordinate Geometry Basics

Q170 Written

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-8, -11)$.

Final answer

$(-8, -11)$

Worked Solutions

Coordinate Geometry Basics

Q171 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0, -15/2)$.

Final answer $(0, -15/2)$

Worked Solutions

Coordinate Geometry Basics

Q172 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 19/2)$.

Correct option: B

Final answer

$(2, 19/2)$

Worked Solutions

Coordinate Geometry Basics

Q173 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(11, -18)$.

Correct option: C

Final answer

$(11, -18)$

Worked Solutions

Coordinate Geometry Basics

Q174 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (10, -15).

Correct option: B

Final answer

(10, -15)

Worked Solutions

Coordinate Geometry Basics

Q175 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8, -11)$.

Correct option: B

Final answer

$(-8, -11)$

Worked Solutions

Coordinate Geometry Basics

Q176 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, -15/2)$.

Correct option: C

Final answer

$(0, -15/2)$

Worked Solutions

Coordinate Geometry Basics

Q177 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3/2, -1/2)$.

Final answer $(3/2, -1/2)$

Worked Solutions

Coordinate Geometry Basics

Q178 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, 2)$.

Final answer $(2, 2)$

Worked Solutions

Coordinate Geometry Basics

Q179 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-1, 0)$.

Final answer $(-1, 0)$

Worked Solutions

Coordinate Geometry Basics

Q180 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2, 1)$.

Final answer $(-2, 1)$

Worked Solutions

Coordinate Geometry Basics

Q181 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (0, 3).

Final answer

(0, 3)

Worked Solutions

Coordinate Geometry Basics

Q182 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2/3, 1)$.

Final answer $(2/3, 1)$

Worked Solutions

Coordinate Geometry Basics

Q183 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, -1)$.

Final answer $(2, -1)$

Worked Solutions

Coordinate Geometry Basics

Q184 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(5/3, 5/3)$.

Final answer $(5/3, 5/3)$

Worked Solutions

Coordinate Geometry Basics

Q185 Written

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, -5/2)$.

Final answer

$(2, -5/2)$

Worked Solutions

Coordinate Geometry Basics

Q186 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0, 0)$.

Final answer $(0, 0)$

Worked Solutions

Coordinate Geometry Basics

Q187 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3/2, -1/2)$.

Correct option: A

Final answer

$(3/2, -1/2)$

Worked Solutions

Coordinate Geometry Basics

Q188 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 2)$.

Correct option: C

Final answer

$(2, 2)$

Worked Solutions

Coordinate Geometry Basics

Q189 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, 0)$.

Correct option: B

Final answer

$(-1, 0)$

Worked Solutions

Coordinate Geometry Basics

Q190 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: A

Final answer

$(-2, 1)$

Worked Solutions

Coordinate Geometry Basics

Q191 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (0, 3).

Correct option: A

Final answer

(0, 3)

Worked Solutions

Coordinate Geometry Basics

Q192 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2/3, 1).

Correct option: D

Final answer

(2/3, 1)

Worked Solutions

Coordinate Geometry Basics

Q193 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, -1)$.

Correct option: B

Final answer

$(2, -1)$

Worked Solutions

Coordinate Geometry Basics

Q194 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(5/3, 5/3)$.

Correct option: A

Final answer

$(5/3, 5/3)$

Worked Solutions

Coordinate Geometry Basics

Q195 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, -5/2)$.

Correct option: B

Final answer

$(2, -5/2)$

Worked Solutions

Coordinate Geometry Basics

Q196 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 0)$.

Correct option: C

Final answer

$(0, 0)$

Worked Solutions

Coordinate Geometry Basics

Q197 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB: find the centroid G**

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(4/3, -2/3)$.

Final answer

$(4/3, -2/3)$

Worked Solutions

Coordinate Geometry Basics

Q198 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB: find the centroid G:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3, -2/3)$.

Correct option: C

Final answer

$(4/3, -2/3)$

Worked Solutions

Coordinate Geometry Basics

Q199 Written

Point with A $(-3, 2)$ symmetric about centre $(2, 6)$

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7, 10)$.

Final answer

$(7, 10)$

Q200 Written

Point with A $(1, -2)$ symmetric about centre $(2, 3)$

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(3, 8)$.

Final answer

$(3, 8)$

Worked Solutions

Coordinate Geometry Basics

Q201 **Written****Point with A (3, 2) symmetric about centre (-1, 5)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-5, 8)$.

Final answer $(-5, 8)$ Q202 **Written****Point with A $(-3, 1)$ symmetric about centre $(2, 0)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7, -1)$.

Final answer $(7, -1)$

Worked Solutions

Coordinate Geometry Basics

Q203 **Written****Point with A (7, 4) symmetric about centre (-3, 5)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-13, 6)$.

Final answer $(-13, 6)$ Q204 **Written****Point with A (4, -8) symmetric about centre (-3, 2)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-10, 12)$.

Final answer $(-10, 12)$

Worked Solutions

Coordinate Geometry Basics

Q205 **Written****Point with A (0, -4) symmetric about centre (-4, 1)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-8, 6)$.

Final answer $(-8, 6)$ Q206 **Written****Point with A $(-4, 2)$ symmetric about centre $(-1, -3)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(2, -8)$.

Final answer $(2, -8)$

Worked Solutions

Coordinate Geometry Basics

Q207 MCQ

Find point with A $(-3, 2)$ symmetric about centre $(2, 6)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(7, 10)$.

Correct option: C**Final answer** $(7, 10)$

Q208 MCQ

Find point with A $(1, -2)$ symmetric about centre $(2, 3)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 8)$.

Correct option: B**Final answer** $(3, 8)$

Worked Solutions

Coordinate Geometry Basics

Q209 MCQ

Find point with A (3, 2) symmetric about centre (-1, 5):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-5, 8).

Correct option: A**Final answer**

(-5, 8)

Q210 MCQ

Find point with A (-3, 1) symmetric about centre (2, 0):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (7, -1).

Correct option: C**Final answer**

(7, -1)

Worked Solutions

Coordinate Geometry Basics

Q211 MCQ

Find point with A (7, 4) symmetric about centre (-3, 5):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-13, 6)$.

Correct option: C**Final answer** $(-13, 6)$

Q212 MCQ

Find point with A (4, -8) symmetric about centre (-3, 2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-10, 12)$.

Correct option: C**Final answer** $(-10, 12)$

Worked Solutions

Coordinate Geometry Basics

Q213 MCQ

Find point with A (0, -4) symmetric about centre (-4, 1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-8, 6).

Correct option: B**Final answer**

(-8, 6)

Q214 MCQ

Find point with A (-4, 2) symmetric about centre (-1, -3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, -8).

Correct option: A**Final answer**

(2, -8)

Worked Solutions

Coordinate Geometry Basics

Q215 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -9 .

Final answer

-9

Worked Solutions

Coordinate Geometry Basics

Q216 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -9 .

Correct option: C

Final answer

-9

Worked Solutions

Coordinate Geometry Basics

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Point A ($-3, 4$) lies in which quadrant?:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A

Final answer

II

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Find the distance between A ($1, 2$) and B ($4, 6$):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: B

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Point P divides the segment from A $(-2, 3)$ to B $(6, 1)$ internally in ratio 1:3. Find P:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 5/2)$.

Correct option: C

Final answer

$(0, 5/2)$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Point B is symmetric to A $(2, -1)$ about P $(-3, 4)$. Find B:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8, 9)$.

Correct option: C

Final answer

$(-8, 9)$

Worked Solutions

Coordinate Geometry Basics

Quiz WRITTEN 5 Concept Quiz · Written

Find the point on $y = 2x + 1$ equidistant from A (0, 2) and B (4, 0).

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (1, 3).

Final answer

(1, 3)

Quiz WRITTEN 6 Concept Quiz · Written

For points A (−5) and B (7), find the point dividing AB externally in ratio 2:5.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives −13.
3. Final answer: −13.

Final answer

−13

Worked Solutions

Coordinate Geometry Basics

Quiz WRITTEN 7 Concept Quiz · Written

If the distance between A (3, -1) and B (x, 2) is 5, find x.

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Quiz WRITTEN 8 Concept Quiz · Written

Find the centroid of A (-2, 1), B (6, -3), C (1, 7).

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (5/3, 5/3).

Final answer

$$(5/3, 5/3)$$

Answer key

Use the question number shown on each page.

Q001 12	Q027 2	Q053 <i>II</i>	Q079 D
Q002 12	Q028 7	Q054 <i>IV</i>	Q080 C
Q003 8	Q029 C	Q055 <i>III</i>	Q081 D
Q004 6	Q030 A	Q056 <i>I</i>	Q082 C
Q005 C	Q031 D	Q057 C	Q083 (3, 2)
Q006 C	Q032 A	Q058 B	Q084 (1, 2)
Q007 C	Q033 5	Q059 A	Q085 (4, -2)
Q008 C	Q034 -7	Q060 C	Q086 (-1, -2)
Q009 4	Q035 -23	Q061 A	Q087 (3, 4)
Q010 3	Q036 -11/2	Q062 C	Q088 C
Q011 4	Q037 A	Q063 B	Q089 C
Q012 8	Q038 C	Q064 A	Q090 A
Q013 B	Q039 B	Q065 B	Q091 A
Q014 C	Q040 C	Q066 A	Q092 A
Q015 B	Q041 <i>I</i>	Q067 A	Q093 (3, -2)
Q016 B	Q042 <i>IV</i>	Q068 C	Q094 (1, -2)
Q017 -28	Q043 <i>II</i>	Q069 B	Q095 (4, 2)
Q018 -13	Q044 <i>III</i>	Q070 C	Q096 (-1, 2)
Q019 18	Q045 <i>II</i>	Q071 C	Q097 (3, -4)
Q020 1	Q046 <i>I</i>	Q072 D	Q098 B
Q021 A	Q047 <i>IV</i>	Q073 (-3, -2)	Q099 B
Q022 B	Q048 <i>III</i>	Q074 (-1, -2)	Q100 B
Q023 A	Q049 <i>II</i>	Q075 (-4, 2)	Q101 A
Q024 B	Q050 <i>IV</i>	Q076 (1, 2)	Q102 D
Q025 2	Q051 <i>I</i>	Q077 (-3, -4)	Q103 (-3, -4)
Q026 1	Q052 <i>III</i>	Q078 C	Q104 (3, 4)

Q105 (3, -4)

Q106 A

Q107 B

Q108 C

Q109 5

Q110 $\sqrt{10}$ Q111 $\sqrt{7}$ Q112 $5\sqrt{5}$ Q113 $4\sqrt{2}$

Q114 C

Q115 A

Q116 C

Q117 B

Q118 A

Q119 $x = -1, 7$ Q120 $x = -2, 0$ Q121 $x = -1, 7$ Q122 $y = -5, -1$ Q123 $x = -14, 6$

Q124 B

Q125 C

Q126 B

Q127 A

Q128 D

Q129 $2\sqrt{5}$

Q130 5

Q131 C

Q132 C

Q133 $x = -2, 3$

Q134 C

Q135 (2, 0)

Q136 (0, 3)

Q137 (2, 0)

Q138 (0, $5/6$)

Q139 (17/2, 0)

Q140 C

Q141 A

Q142 C

Q143 C

Q144 B

Q145 (13/8, 9/4)

Q146 (-4, -11)

Q147 (2, 1)

Q148 (3, 2)

Q149 (-2, 3)

Q150 B

Q151 C

Q152 A

Q153 C

Q154 B

Q155 $3\sqrt{5}$

Q156 A

Q157 (4/3, 11/3)

Q158 (-15/4, 13/4)

Q159 (2, -9/5)

Q160 (3, 3)

Q161 (-2/3, -5/3)

Q162 C

Q163 C

Q164 C

Q165 C

Q166 A

Q167 (2, 19/2)

Q168 (11, -18)

Q169 (10, -15)

Q170 (-8, -11)

Q171 (0, -15/2)

Q172 B

Q173 C

Q174 B

Q175 B

Q176 C

Q177 (3/2, -1/2)

Q178 (2, 2)

Q179 (-1, 0)

Q180 (-2, 1)

Q181 (0, 3)

Q182 (2/3, 1)

Q183 (2, -1)

Q184 (5/3, 5/3)

Q185 (2, -5/2)

Q186 (0, 0)

Q187 A

Q188 C

Q189 B

Q190 A

Q191 A

Q192 D

Q193 B

Q194 A

Q195 B

Q196 C

Q197 (4/3, -2/3)

Q198 C

Q199 (7, 10)

Q200 (3, 8)

Q201 (-5, 8)

Q202 (7, -1)

Q203 (-13, 6)

Q204 (-10, 12)

Q205 (-8, 6)

Q206 (2, -8)

Q207 C

Q208 B

Q209 A

Q210 C

Q211 C

Q212 C

Q213 B

Q214 A

Q215 -9

Q216 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 C

Quiz WRITTEN 5 (1, 3)

Quiz WRITTEN 6 -13

Quiz WRITTEN 7 $x = -1, 7$

Quiz WRITTEN 8 (5/3, 5/3)